Magnetism

Exam code: 0625 & 0972

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Magnetism

Forces between magnetic poles

- The ends of a magnet are called **poles**
- Magnets have two poles: a **north** and a **south**
- Magnetic forces are **strongest** at the poles

North and south poles of a magnet

Poles of a Magnet

- When two magnets are held close together, there will be a force between the magnets
  - **Like poles repel** (push each other apart)
    - A north pole will repel a north pole
    - A south pole will repel a south pole
  - **Opposite poles attract** (are pulled toward each other)
    - A north pole will be attracted to a south pole

Magnet attraction and repulsion

Opposite poles attract and like poles repel

- The attraction or repulsion between two magnetic poles is an example of a **non-contact force**

Help me understand

Stuck on something?

Ask questions and get help with exactly what you're revising.

Got it

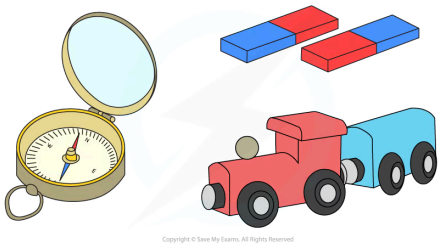
Examiner Tips and Tricks

Uses of magnets

Uses of permanent magnets

- Uses of permanent magnets include:
  - **Compasses:** for thousands of years humans have used compasses for navigation, since the needle always points north
  - **School lab experiments:** the magnets used in school science demonstrations are permanent magnets
  - **Toys:** toy trains and trucks often have magnets which attach the carriages or trailers to the engine or cab
  - **Fridge magnets:** these are made either of flexible magnetic material or by sticking a magnet to the back of something

Some uses of magnets

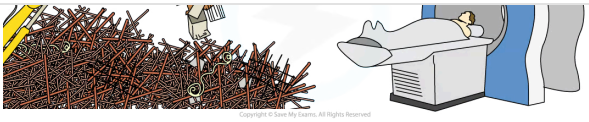


Compasses and toy trains are two examples of the use of permanent magnets

Uses of electromagnets

- Electromagnets have many uses including
  - **MRI scanners:** in hospitals, an MRI scanner is a large, cylindrical machine using powerful electromagnets to produce diagnostic images of the organs of the body
  - **Speakers and earphones:** the loudspeakers, microphones and earphones used in phones and laptops use electromagnets to sense or send soundwaves
  - **Recycling:** because steel is a magnetic material it can be easily separated from other metals and materials using electromagnets. Once recovered the steel is re-used and recycled, reducing mining for iron ore and processing ore into steel
  - **Mag-Lev Trains:** the ability of Mag-Lev trains to hover above the rails is due to them being repelled by large electromagnets on the train and track. This reduces friction and allows speeds of nearly 400 miles per hour

Some uses of electromagnets



Picking and sorting scrap metal and an MRI scanner are two examples of the uses of electromagnets

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